



The Recommender System for Big Data

In social networking sites and in e-commerce, recommendations play a key role. E-commerce, in particular, depends on website traffic which is driven to it by various means. Fortunately, machine learning algorithms like Recommender can be used effectively in both these cases.

This article focuses on Mahout's recommender systems.

Mahout is an open source machine learning library from the Apache Software Foundation. It implements many data mining algorithms like *recommender engines()*, *clustering ()*, *classification ()*, and is scalable to very large data sets—up to terabytes and petabytes, which is in the Big Data realm.

In this article, I will focus on recommender systems in Mahout. Recommender engines are very popular machine learning algorithms that are used to recommend books, movies or articles based on the users' past actions and interests.

Recommender engine algorithms come in two categories – user based and item based recommendations. User based recommendations predict what the users will like, based on their similarity with other users and do not require the properties or description of the items; for example, Facebook's friend recommendations. In item based recommendations, the preferences of users for some items based on their preference of other similar items can be suggested, and this requires prior knowledge of the particular item's properties. For example, if a user likes one kind of comedy movie, he may like other similar comedy movies.

The following are the main components of the recommendation system in Mahout.

- **Data model interface:** This converts the raw data to a Mahout compliant data format. Data sources like MySQL, PostgreSQL, MongoDB, Cassandra, flat files, etc, are supported.
- **User similarity interface:** This contains methods to compute the similarity among users.
- **Item similarity interface:** This contains methods to

compute the similarity between items. Many similarity metrics are available in Mahout like Pearson Correlation, Euclidean distance, etc.

- **User neighbourhood interface:** This is a method to construct a neighbourhood around a given user, which satisfies the similarity or 'nearest neighbour' threshold criteria.
- **Recommender interface:** This is a method that finally makes the actual recommendation.

In order to process Big Data, Mahout can use HDFS. So a prerequisite to installing Mahout is JDK/JAVA 1.7 or later, Maven 3.0 or later and the Hadoop cluster. If you are using Ubuntu, you can use the following commands.

Install Maven from the repository `sudo apt-get install maven`. Download the latest distribution of Mahout from the site <http://www.apache.org/dyn/closer.cgi/lucene/mahout/>. Unzip and copy the following command to the desired location:

```
cp -R SOURCE DESTINATION
```

I copied it in `/usr/local`.

Now, `cd /usr/local/mahout/distribution` or wherever you have copied Mahout. Then run the following command:

```
sudo mvn install
```

(Install Maven 3.0.1 or above for Mahout .20 distribution; else, it will throw some error.) You will get the screen shown in Figure 1 after successful installation.

Mahout's recommenders expect interactions between users and items as input. I tested a sample 568.3MB data,